

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions

- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|,$$

extracted via **emtrends**. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff’s δ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in analysis/human-ai/novelty.Rmd and is not duplicated here. Starter-mechanism analysis is deferred.

Article-use note. The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	80	640
human-ai	96	667
human-human	36	288

3 Preprocessing

Slot-level rows: 3034

Metric rows: 9102

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from $\text{resonance} \sim \text{surprise} *$ slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	1.091	1.048	-0.043	0.038	-1.152	0.249	0.043	
Human-Human	1.063	1.052	-0.012	0.054	-0.215	0.830	0.012	
AI-AI	0.961	1.085	0.124	0.031	3.941	0.000	0.124	***

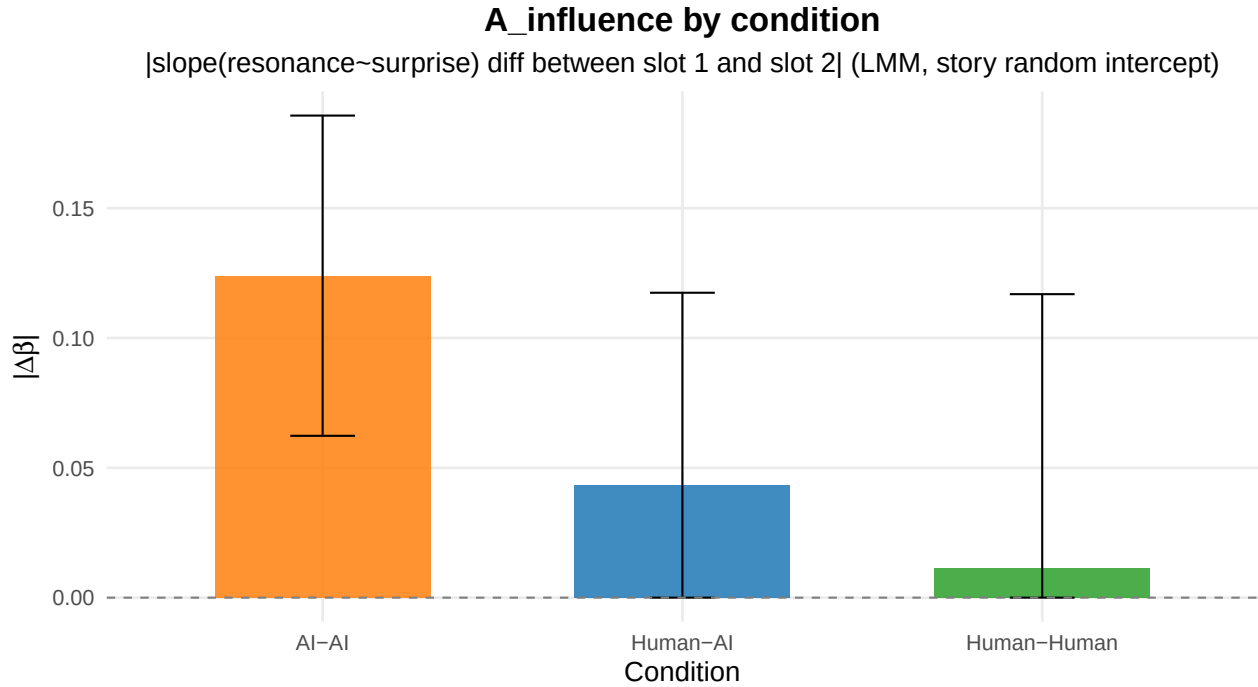


Table 3: Pooled model: |slope_slot1 - slope_slot2| (unsigned) by condition

condition	A_influence	se
Human-AI	0.029	0.037
Human-Human	0.014	0.053
AI-AI	0.133	0.034

Table 4: Robustness: per-story |slope_1 - slope_2| (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	96	0.400	0.475
Human-Human	36	0.286	0.373
AI-AI	80	0.293	0.316

```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 9.3689, df = 2, p-value = 0.009238
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
```

```
## data: per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
##           Human-AI Human-Human
## Human-Human 0.2841    -
## AI-AI       0.0071    0.2841
##
## P value adjustment method: BH
```

5 Bootstrap CIs and Effect Sizes on A_influence_story

Table 5: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	96	0.475	0.410	0.546
Human-Human	36	0.373	0.297	0.455
AI-AI	80	0.316	0.269	0.367

Table 6: Pairwise Cliff’s delta on per-story A_influence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	96	36	0.146	negligible
Human-AI vs AI-AI	96	80	0.267	small
Human-Human vs AI-AI	36	80	0.125	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: per-story A_influence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.13	0.042	3.071	154.173	0.003	0.047	0.214

7 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned A_influence is similar.

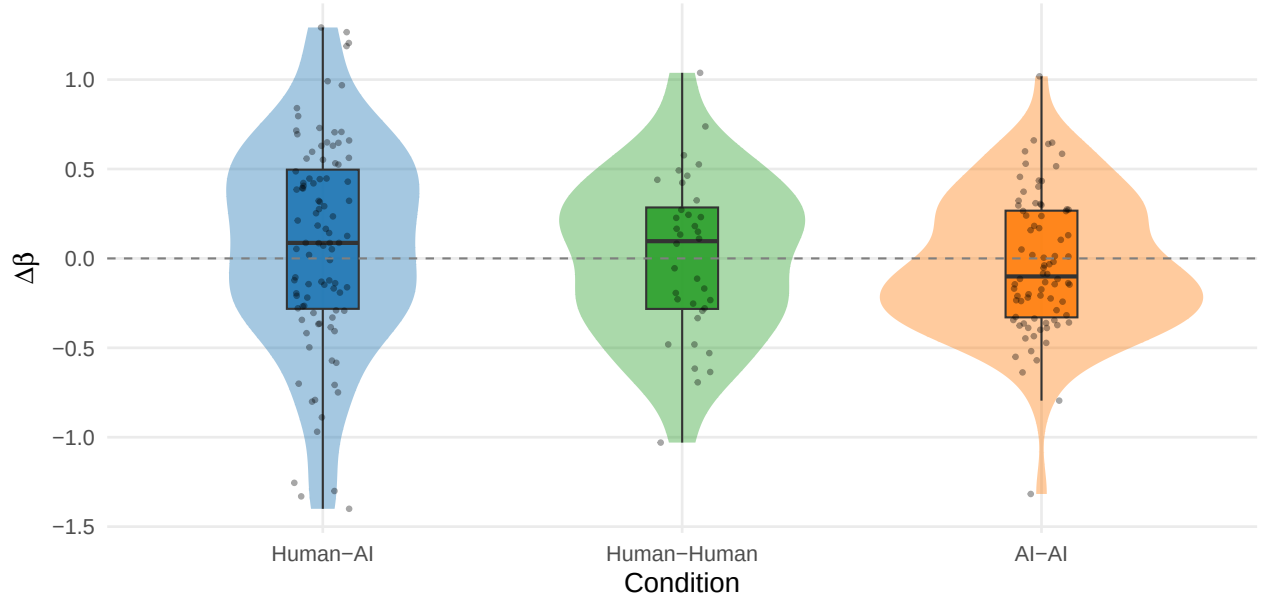
Table 8: Per-condition directional stability of signed novelty-resonance asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	96	0.081	0.582	0.139	0.573	0.177
Human-Human	36	0.006	0.451	0.012	0.528	0.942
AI-AI	80	-0.036	0.389	0.092	0.575	0.413

Brown-Forsythe variance-equality p = 0.0025

Signed novelty-resonance slope asymmetry by condition

slope_slot1 - slope_slot2 per story



8 Length-Controlled Variant of A_influence

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

Table 9: Length-controlled $|\text{slope_slot1} - \text{slope_slot2}|$ (resonance $\sim \text{surprise} * \text{slot} + \text{nwords}$)

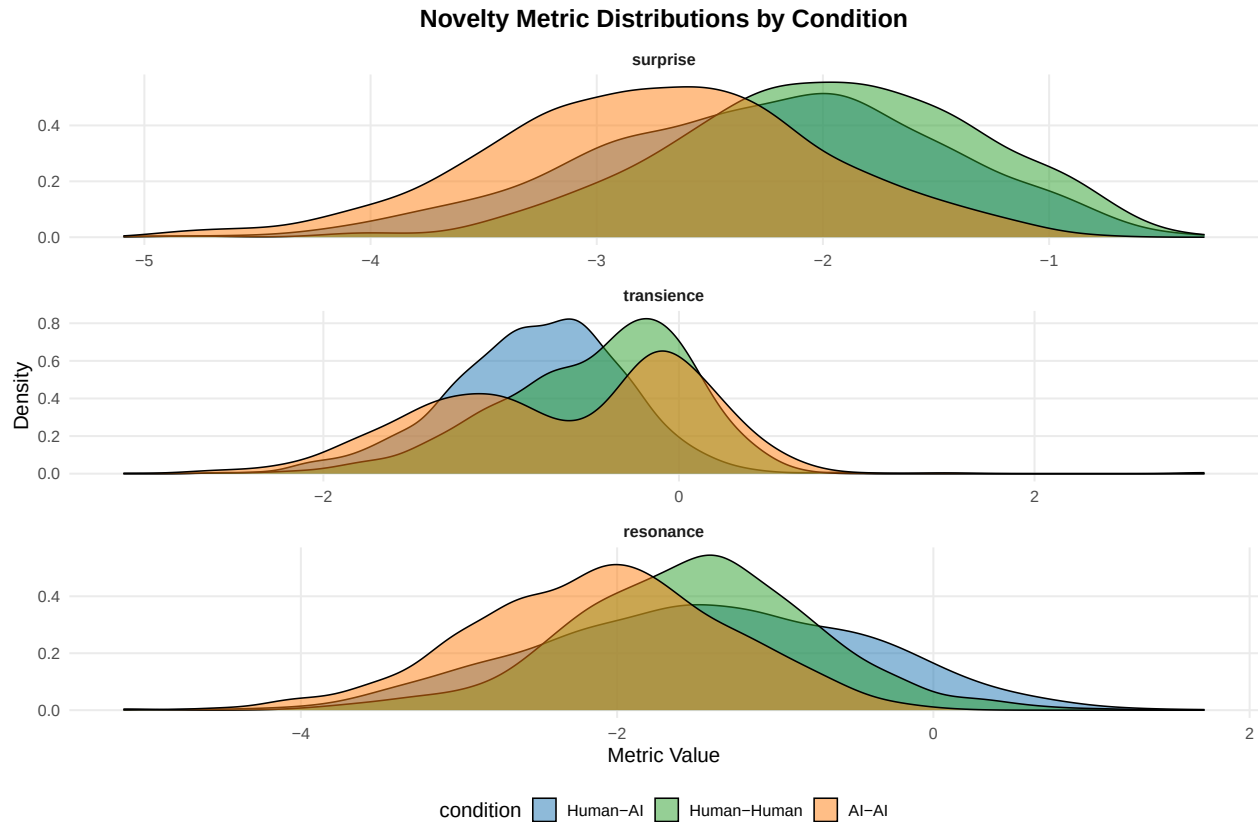
condition	A_influence_lc	se	p
Human-Human	0.012	0.054	0.828

9 Supplemental: General Condition-Level Comparison

Table 10: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	96	1238	-2.264	0.772

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	transience	96	1238	-0.807	0.490
Human-AI	resonance	96	1238	-1.457	1.020
Human-Human	surprise	36	557	-1.992	0.675
Human-Human	transience	36	557	-0.467	0.545
Human-Human	resonance	36	557	-1.526	0.787
AI-AI	surprise	80	1239	-2.746	0.719
AI-AI	transience	80	1239	-0.635	0.729
AI-AI	resonance	80	1239	-2.111	0.798



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6406.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9249 -0.6242  0.0259  0.6727  2.9950
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 0.1020   0.3194
```

```

## Residual                      0.4351    0.6596
## Number of obs: 3034, groups:  conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -2.26528    0.03761 222.05851 -60.227 < 2e-16 ***
## conditionHuman-Human    0.26733    0.07092 208.88531   3.769 0.000213 ***
## conditionAI-AI      -0.48789    0.05515 212.09573  -8.847 3.56e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.530
## condtnAI-AI -0.682  0.362

## contrast      estimate      SE df z.ratio p.value
## (Human-AI) - (Human-Human)   -0.267 0.0709 Inf  -3.769  0.0005
## (Human-AI) - (AI-AI)         0.488 0.0551 Inf   8.847  <.0001
## (Human-Human) - (AI-AI)      0.755 0.0724 Inf  10.431  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4027.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.1148 -0.5719  0.0334  0.5949  8.6129
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.1783  0.4222
## Residual              0.1822  0.4269
## Number of obs: 3034, groups:  conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -0.80625    0.04477 213.90153 -18.008 < 2e-16 ***
## conditionHuman-Human    0.32897    0.08535 210.09344   3.855 0.000154 ***
## conditionAI-AI      0.15299    0.06618 211.02854   2.312 0.021766 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.525
## condtnAI-AI -0.676  0.355

```

```

## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -0.329 0.0853 Inf -3.855 0.0003
## (Human-AI) - (AI-AI) -0.153 0.0662 Inf -2.312 0.0624
## (Human-Human) - (AI-AI) 0.176 0.0875 Inf 2.011 0.1328
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 7859
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.2695 -0.6260 0.0110 0.6547 3.7172
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.06171 0.2484
## Residual 0.73636 0.8581
## Number of obs: 3034, groups: conversation_id, 212
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -1.45840 0.03519 237.64894 -41.446 <2e-16 ***
## conditionHuman-Human -0.06495 0.06538 210.92213 -0.993 0.322
## conditionAI-AI -0.64664 0.05103 217.37061 -12.671 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H
## cndtnHmn-Hm -0.538
## condtnAI-AI -0.690 0.371
##
## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) 0.065 0.0654 Inf 0.993 0.9616
## (Human-AI) - (AI-AI) 0.647 0.0510 Inf 12.671 <.0001
## (Human-Human) - (AI-AI) 0.582 0.0664 Inf 8.766 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests

```

10 Supplemental: Pooled Novelty-to-Resonance Relationship

```

## Novelty-to-resonance model:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)

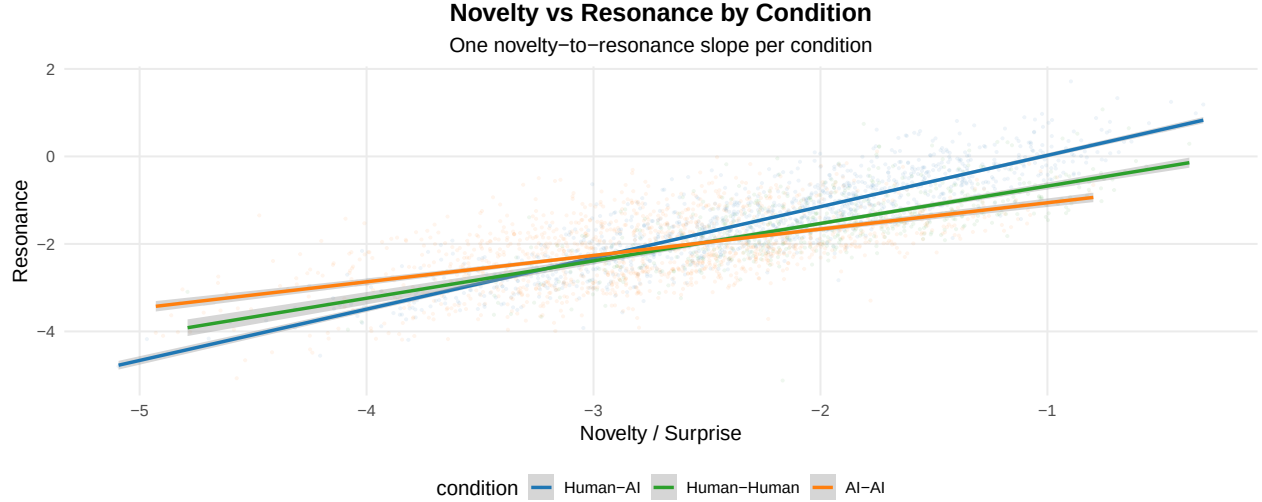
```



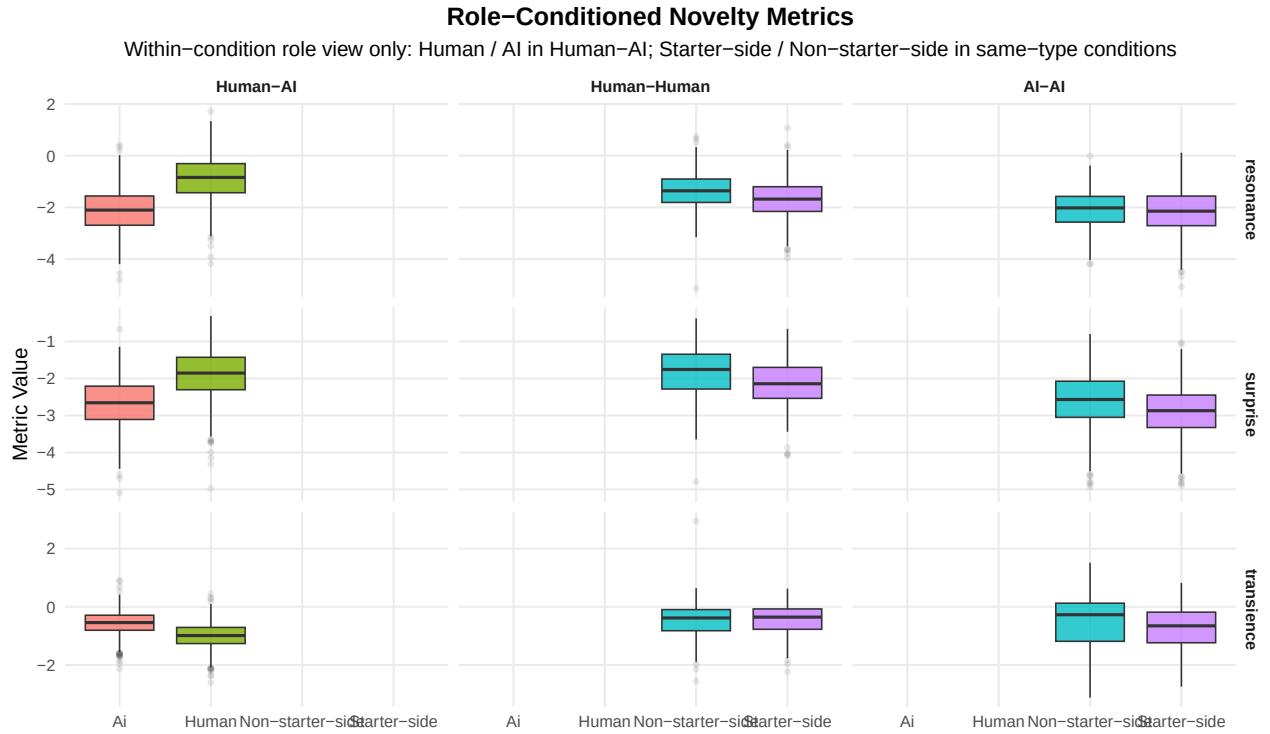
```

## Data: df_slots
##
## REML criterion at convergence: 3822.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.9603 -0.5885 -0.0344  0.5463  4.1918
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.1828    0.4275
## Residual                   0.1684    0.4103
## Number of obs: 3034, groups: conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      1.35712    0.05824  547.44955  23.304 < 2e-16
## surprise          1.24314    0.01622 2873.02810  76.626 < 2e-16
## conditionHuman-Human -0.76827    0.11124  537.45432  -6.906 1.41e-11
## conditionAI-AI      -0.80809    0.09388  718.28978  -8.607 < 2e-16
## surprise:conditionHuman-Human -0.18736    0.03413 2911.35544  -5.489 4.38e-08
## surprise:conditionAI-AI -0.28102    0.02567 2927.03894 -10.949 < 2e-16
##
## (Intercept)          ***
## surprise              ***
## conditionHuman-Human ***
## conditionAI-AI        ***
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise      0.631
## cndtnHmn-Hm   -0.523 -0.330
## condtnAI-AI   -0.620 -0.392  0.325
## srprs:cnH-H   -0.300 -0.475  0.632  0.186
## srprs:AI-AI   -0.399 -0.632  0.209  0.700  0.300
##
## Condition-specific novelty-to-resonance slopes:
## condition surprise.trend SE df asymp.LCL asymp.UCL
## Human-AI      1.243 0.0162 Inf  1.211 1.27
## Human-Human    1.056 0.0300 Inf  0.997 1.11
## AI-AI          0.962 0.0199 Inf  0.923 1.00
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95

```



11 Supplemental: Role-Level View



12 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from $\text{resonance} \sim \text{surprise} * \text{slot} + (1|\text{conversation_id})$. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story $A_{\text{influence}}$
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

- length-controlled re-fit (+ `nwords`) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled `condition` LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in `analysis/human-ai/novelty.Rmd`)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-10 21:18:28.747882